

Summary of Proposed ISAC-based UAV System Model

1 Signal Transmission Process

The proposed system is based on an Orthogonal Frequency Division Multiplexing (OFDM) based Integrated Sensing and Communication (ISAC) framework.

Step 1: The Base Station (BS), equipped with multiple antennas, generates an OFDM waveform.

Step 2: The BS performs transmit beamforming to simultaneously:

- communicate with the mobile user,
- illuminate the sensing targets (vehicle and eavesdroppers).

Step 3: A passive RIS-mounted UAV receives the incident OFDM waveform from the BS and intelligently reflects it towards the mobile user by adjusting the phase shifts of its RIS elements.

Step 4: The RIS-mounted UAV operates in half-duplex mode and employs a passive Reconfigurable Intelligent Surface (RIS) consisting of a 4×4 planar array (16 reflecting elements) to enhance communication coverage and signal quality.

Step 5: The mobile user follows a Random Walk mobility model.

Step 6: Simultaneously, the BS illumination reaches the vehicle and HPPP-distributed eavesdroppers. Their reflected signals form the **bistatic sensing path**.

Step 7: A Full-Duplex UAV Jammer simultaneously transmits jamming signals while receiving reflected echoes.

Step 8: The jammer's transmitted waveform is reflected from the vehicle and eavesdroppers, producing the **monostatic sensing path**.

Step 9: The UAV Jammer receives two independent reflected copies:

- Bistatic reflection (BS $\rightarrow$ Target $\rightarrow$ Jammer)
- Monostatic reflection (Jammer $\rightarrow$ Target $\rightarrow$ Jammer)

Step 10: Selection Combining (SC) is employed at the jammer. The reflected signal having the higher instantaneous SNR is selected for sensing.

Step 11: The selected echo is used for target sensing and localization while the jammer simultaneously degrades the eavesdroppers' ability to intercept the communication.

2 System Model and Components

The proposed ISAC system consists of the following entities.

Base Station (BS)

- Multiple antennas
- OFDM waveform generation
- Transmit beamforming
- Communication transmitter
- Bistatic radar illuminator

Mobile User

- Receives communication data
- Mobility modeled using Random Walk

RIS-mounted UAV

- Half-Duplex operation
- Passive Reconfigurable Intelligent Surface (RIS)
- Assists BS–User communication through intelligent signal reflection
- 16 reflecting elements (4×4 planar array)
- Adaptive RIS phase-shift control

Full-Duplex UAV Jammer

- Multiple antennas
- MISO transmission
- Simultaneous jamming and sensing
- Monostatic radar operation
- Receives bistatic and monostatic echoes
- Performs Selection Combining

Moving Vehicle

- Passive sensing target
- Reflects BS and jammer signals

Eavesdroppers

- Spatially distributed according to a Homogeneous Poisson Point Process (HPPP)
- Intercept communication
- Reflect radar signals
- Are jammed by the UAV jammer

Optimization Objective

The system focuses primarily on sensing performance while maintaining secure communication.

Main objectives include:

- Minimize Cramér–Rao Bound (CRB)
- Maximize Detection Probability (P_D)
- Maximize Secrecy Rate (R_s)

The overall optimization is expected to be solved using a Multi-Agent Deep Reinforcement Learning (MADRL) algorithm due to the dynamic environment introduced by user mobility and randomly distributed eavesdroppers.

3 Channel Model

For simplicity, every wireless link follows the same channel model.

Small-scale fading

All wireless channels are modeled using the Rician fading model,

$$h = \sqrt{\frac{K}{K+1}} h_{\text{LOS}} + \sqrt{\frac{1}{K+1}} h_{\text{NLOS}},$$

where

- K is the Rician factor,
- h_{LOS} represents the deterministic Line-of-Sight component,
- h_{NLOS} represents the scattered multipath component.

The same Rician model is assumed for all communication and sensing links for simplicity.

Examples include

- BS $\leftrightarrow$ RIS-mounted UAV

- RIS-mounted UAV $\leftrightarrow$ User
- BS $\leftrightarrow$ Vehicle
- BS $\leftrightarrow$ Eavesdroppers
- Jammer $\leftrightarrow$ Vehicle
- Jammer $\leftrightarrow$ Eavesdroppers

Large-scale fading

Large-scale fading is modeled using an elevation-angle dependent Air-to-Ground (A2G) path-loss model.

The elevation angle is

$$\theta = \tan^{-1} \left(\frac{H}{d} \right),$$

where

- H is UAV altitude,
- d is horizontal distance.

The path loss is therefore a function of the elevation angle and captures LOS probability more accurately than conventional terrestrial path-loss models.

4 Passive RIS Model

The communication UAV is equipped with a passive Reconfigurable Intelligent Surface (RIS), which intelligently reflects the incident BS signal towards the intended mobile user without performing active signal regeneration or relaying.

The RIS consists of

- 16 passive reflecting elements,
- 4×4 planar array configuration.

The RIS reflection coefficient matrix is

$$\mathbf{\Phi} = \text{diag}(\beta_1 e^{j\phi_1}, \beta_2 e^{j\phi_2}, \dots, \beta_{16} e^{j\phi_{16}}),$$

where

- β_n denotes the reflection amplitude,
- ϕ_n denotes the phase shift of the n^{th} RIS element.

For an ideal passive RIS,

$$\beta_n = 1,$$

thus only the phase shifts are optimized.

5 Sensing Performance Metrics

Since the proposed work focuses primarily on sensing, the following sensing metrics are considered.

1. Cramér-Rao Bound (CRB)

CRB provides the theoretical lower bound on the variance of any unbiased estimator.

A lower CRB indicates higher sensing accuracy and better target parameter estimation.

Optimization objective:

Minimize CRB

2. Detection Probability (P_D)

Detection Probability measures the likelihood of correctly detecting the presence of a target.

A larger value of P_D indicates better sensing capability.

Optimization objective:

Maximize P_D

6 Communication Performance Metric

Although the primary focus of the proposed work is sensing, communication performance must also be evaluated to analyze the sensing–communication tradeoff.

Secrecy Rate (R_s)

Since the system contains randomly distributed eavesdroppers, Physical Layer Security (PLS) is incorporated by employing a Full-Duplex UAV Jammer. The Full-Duplex UAV Jammer transmits artificial interference towards the eavesdroppers while the RIS-mounted UAV enhances the legitimate BS–User communication link. Consequently, the secrecy rate quantifies the communication security achieved alongside sensing performance.

The secrecy rate is defined as

$$R_s = [R_b - R_e]^+,$$

where

- R_b denotes the achievable rate of the legitimate communication link,
- R_e denotes the achievable rate of the eavesdropper,
- $[x]^+ = \max(x, 0)$.

A larger secrecy rate indicates stronger communication security against eavesdropping attacks.

Optimization objective:

$$\boxed{\text{Maximize } R_s}$$

The secrecy rate will be jointly analyzed with the sensing metrics (CRB and Detection Probability) to evaluate the sensing–communication tradeoff of the proposed ISAC system.